

Alden Sio

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EDUCATION

National University Of Singapore

Singapore

Bachelor of Science in Business Analytics (Computing)

Aug. 2022 – Apr. 2026

Teaching Assistant: CS2030 (Programming Methodology II), BT1101 (Introduction to Business Analytics)

WORK EXPERIENCE

Tiktok

Singapore

Machine Learning Engineer

Jan. 2025 - Present

- Architected **Agentic Deep Research** using internal & external tools for automated report generations, reducing manual editing efforts by **20 hours weekly**
- Established scalable frameworks for **data ingestion, reuse, and synthetic generation**, building comprehensive asset libraries with natural language and DSL-based search to improve model training diversity
- Built a **Topic Modeling** pipeline using **BERTopic** and various in-house tools, processing over **4 million videos** and parallelizing workflows to achieve a **70%** reduction in runtime
- Configured an **Attribution Model** to guide Shop-Ads Creative clients on optimizing video storytelling, content strategies, and influencer collaborations, leveraging insights from video engagement and CTR

Synapse

Singapore

AI Engineer

May. 2024 - Oct. 2024

- Pioneered a **Self-RAG** and **RAG-Fusion** system using **AWS Bedrock**, boosting response relevancy by **60%** and slashing hallucination rates from **40%** to **2%**
- Engineered a multi-modal data ingestion pipeline with **Kedro**, automating processing of text, tables, and images to save over **150 annual hours** in manual effort
- Integrated **GraphRAG** for extensive context understanding, constructing a **50k** entity knowledge graph from unstructured data

Panda Commercial Bank PLC.

Phnom Penh, Cambodia

Software Engineer

May. 2023 – Aug. 2023

- Implemented and deployed scalable microservices overseeing **e-Wallet transactions** & **e-KYC**
- Devised Data Pipelines for Data Cleaning, Extraction, & Analysis, reducing development time by **25%**

PROJECTS

Agent Visualiser ([GitHub](#)) | *LangChain, LangGraph, Milvus, Neo4j, Chainlit*

Sep. 2024 – Oct. 2024

- Built an E2E Agentic-Hybrid-RAG Chatbot to help my colleagues visualise Agentic Interactions
- Accelerated team productivity by **30%** by streamlining knowledge transfer and reducing learning curves

FashoReco ([GitHub](#)) | *pandas, sklearn, word2vec, scaling, clustering*

Jan. 2024 – May. 2024

- Developed an ensemble Recommendation System using a Time-Infused Collaborative Model, and a Time-Decay Content Based Model, improving recommendation hit rate from **0.03%** to **0.20%**
- Employed Multi-Dimensional Clustering to facilitate in-depth Network Analysis for tailored recommendations

InTurn ([Website](#)) | *HTML, CSS, Vue, JavaScript, Firebase, RESTful APIs*

Jan. 2024 – May. 2024

- Created an all-in-one internship tracking Web-app, saving sourcing overhead time by a week
- Managed and led a team of 5 developers to complete the project a month earlier than planned

TECHNICAL SKILLS

Programming Languages: Python, Java, R, Tableau, SQL, JavaScript, HTML, CSS

Developer Tools: AWS, Git, Postman, Firebase, Google Cloud Platform, Docker

Libraries/Packages: Vue, PyTorch, React Native, langchain, langgraph, FastAPI, LlamaIndex, Neo4j